

TASK 3: Architecture Design

Smart City – Architecture Comparison

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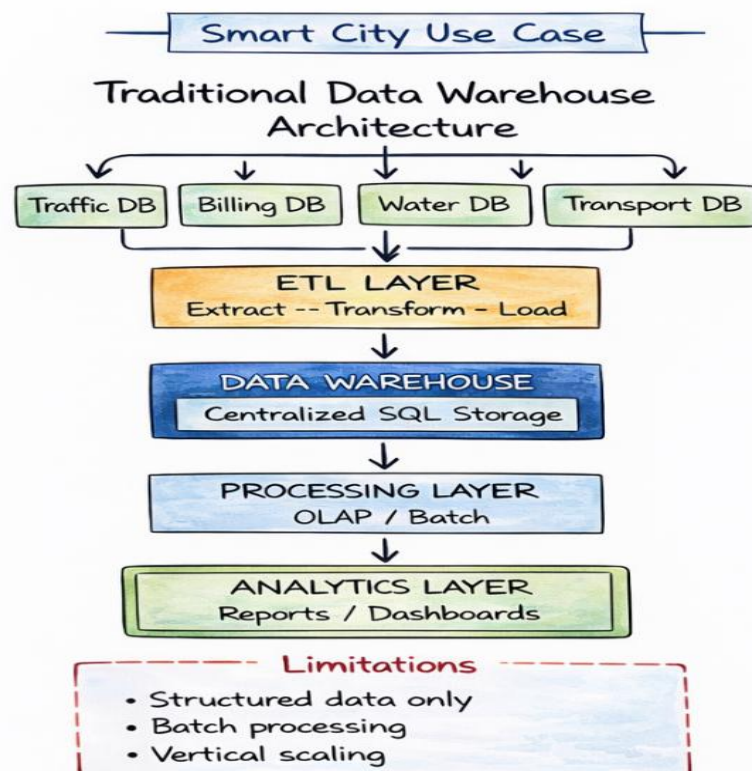
Course: Big Data Analytics

Architecture of Smart City

Traditional Data Warehouse:

A Traditional Data Warehouse architecture is a centralized data management system designed to store and analyze structured data collected from multiple Smart City operational systems such as traffic databases, electricity billing systems, water management records, and public transport systems. It primarily supports Business Intelligence (BI) activities including reporting, dashboard generation, and historical data analysis.

This architecture mainly supports descriptive analytics and batch processing. However, it has limitations in handling unstructured data such as CCTV footage and social media feedback, real-time streaming data from IoT sensors, and large-scale distributed workloads. It also relies on vertical scaling, which restricts flexibility and scalability.



Hadoop-Based Big Data Architecture:

A Hadoop-Based Big Data Architecture is a distributed data processing framework designed to handle large volumes of structured, semi-structured, and unstructured data generated in a Smart City environment, such as CCTV footage, IoT sensors, GPS tracking, and social media data. It stores data in HDFS and processes it using distributed technologies like Spark and MapReduce.

This architecture supports real-time analytics, parallel processing, and horizontal scalability. It enables faster decision-making, predictive analysis, and efficient management of city operations.

